We aren’t quite at the point where we can simply say, “Yes, this emergent geometry on space evolves with time in exactly the right way to describe a spacetime that obeys Einstein’s equation of general relativity.” That’s the ultimate goal, but we’re not there yet. What we can do is to specify a list of requirements under which that’s exactly what does happen. The individual requirements seem reasonable—things like “at long distances, physics looks like an effective quantum field theory”—but many of them remain unproven as yet, and so far the most rigorous results are available only in situations where the gravitational field is relatively weak. We don’t yet have a way of describing black holes or the Big Bang, though there are some promising ideas.

That’s life as a theoretical physicist. We don’t have all the answers, but let’s not lose sight of the overall ambition: starting from an abstract quantum wave function, we have a road map describing how space emerges, with a geometry fixed by quantum entanglement, and that geometry seems to obey the dynamical rules of general relativity. There are so many caveats and assumptions going into this proposal that it’s hard to know where to start listing them. But there seems to be a very real prospect that the route to understanding the universe lies not in quantizing gravity, but in finding gravity within quantum mechanics.

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You may have noticed a tiny imbalance in this discussion. We’ve been asking how spacetime can emerge from entanglement in quantum gravity. But if we’re honest, we’ve really only looked at how space emerges; we’ve taken time for granted as something that comes along for the ride. And it’s possible that this approach is completely fair. Although relativity treats space and time as if they were on an equal footing, quantum mechanics generally does not. The Schrödinger equation, in particular, treats them very differently: it literally describes how the quantum state evolves with time. “Space” may or may not be part of that equation, depending on what system we’re looking at, but time is fundamental. It’s plausible that the symmetry between space and time that we’re familiar with from relativity isn’t built into quantum gravity, but emerges in the classical approximation.